A quadratic inverse-branch Ruelle operator: exact stability traces and a scoped primitive product

Hénon Dynamics Structure Program

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Abstract

For $F(z) = z^2 - 6$ we construct two strict inverse branches on $\mathbb{D}_4$ and a trace-class weighted composition operator on $H^2(\mathbb{D}_4)$. Every periodic point is exhausted by inverse words. The weights $m = 0, 1$ collapse exactly, whereas $m = 2$ gives nontrivial multiplier traces, six exact coefficients, and a primitive product beginning at stability index $k = 2$. The determinant is entire; the displayed raw product is claimed only in its proved disk $|u| < 4$. This is a source-side benchmark, not a target-divisor or quantization result.

Frozen model. Put $\mathbb{D}_4 = \{z : |z| < 4\}$. The disk $6 + \mathbb{D}_4 = D(6, 4)$ is contained in the right half-plane, so its principal square root defines

$$\psi_\pm(z) = \pm\sqrt{z+6}, \qquad (\mathcal{L}_m f)(z) = \sum_{\epsilon=\pm} (\psi'_\epsilon(z))^m f(\psi_\epsilon(z)), \quad m = 0, 1, 2.$$

The owner is $H^2(\mathbb{D}_4)$ with $e_j(z) = (z/4)^j$; one branch is one clock step and $D_2(u) = \det(I - u\mathcal{L}_2)$.

Theorem 1. *Each $\mathcal{L}_m$ is trace class. For every $n \geq 1$, the 2^n roots of $F^n(z) - z$ are simple, lie in $\mathbb{D}_4$, and are the fixed points of the 2^n rooted inverse words. Writing $\Lambda_n(p) = (F^n)'(p)$,*

$$\operatorname{Tr} \mathcal{L}_m^n = \sum_{F^n(p)=p} \frac{\Lambda_n(p)^{-m}}{1 - \Lambda_n(p)^{-1}}. \tag{1}$$

In particular $\operatorname{Tr} \mathcal{L}_0^n = 2^n$, $\operatorname{Tr} \mathcal{L}_1^n = 0$, *and*

$$\operatorname{Tr} \mathcal{L}_2^n = \sum_{F^n(p)=p} \frac{1}{\Lambda_n(p)(\Lambda_n(p) - 1)}.$$

Thus $\det(I - u\mathcal{L}_0) = 1 - 2u$, $\det(I - u\mathcal{L}_1) = 1$, *and D_2 is entire. For $|u| < 4$ it also has the absolutely convergent product*

$$D_2(u) = \prod_{[p] \text{ primitive}} \prod_{k=2}^{\infty} \left(1 - u^{\ell(p)} \Lambda_p^{-k}\right). \tag{2}$$

No convergence of the raw product (2) is asserted outside $|u| < 4$.

Proof. On $\overline{\mathbb{D}}_4$, $2 \leq |z + 6| \leq 10$, hence $|\psi_\pm(z)| \leq \sqrt{10} < 4$ and $|\psi'_\pm(z)| \leq q = 1/(2\sqrt{2})$. The principal-root formula also gives $\Re\psi_+ \geq \sqrt{2}$ and $\Re\psi_- \leq -\sqrt{2}$. For $m = 2$,

$$M_{(\psi'_\epsilon)^2} C_{\psi_\epsilon} = \sum_{j \geq 0} \left[(\psi'_\epsilon)^2 (\psi_\epsilon/4)^j\right] \otimes e_j^*.$$

The coefficient functionals have norm one and bounded holomorphic functions have Hardy norm at most their supremum. Therefore

$$\|\mathcal{L}_2\|_1 \leq 2\frac{1/8}{1 - \sqrt{10}/4} = \frac{1}{4 - \sqrt{10}}.$$

The same geometric expansion, with the corresponding bounded weights, proves trace class for $m = 0, 1$.

If $|z| > 3$, then $|F(z)| \geq |z|^2 - 6 > |z|$, so every periodic point lies in $\mathbb{D}_4$. Each length-n inverse word is a q^n contraction into the interior and has one fixed point. Conversely, a periodic point chooses one sign at each inverse step, uniquely because the branch images are separated. These points exhaust the monic degree-2^n polynomial $F^n - z$. At the fixed point attached to a word w, the chain rule gives $\psi_w'(p) = \Lambda_n(p)^{-1}$; its modulus is at most $q^n < 1$, so no root is multiple.

For a strict disk map ϕ, holomorphic weight g, and fixed point a, conjugating a to zero makes $M_g C_\phi$ triangular, with diagonal $g(a)\phi'(a)^j$. Its nuclear trace is $g(a)/(1-\phi'(a))$. Expanding $\mathcal{L}_m^n$ over inverse words and applying the chain rule proves (1).

Let $P_n = F^n - z$. It is monic of degree 2^n, $P_n'(p) = \Lambda_n(p) - 1$, and Lagrange interpolation gives $\sum_{P_n(p)=0} 1/P_n'(p) = 0$. Equation (1) now gives the $m = 0, 1$ formulas. Their determinant identities follow first near zero from the trace logarithm and then globally by entireness.

For a primitive orbit put $\mu_p = \Lambda_p^{-1}$. In the rooted trace at time $n = r\ell(p)$ that orbit occurs at each of its $\ell(p)$ points; this cancels the same factor in $1/n$. Regrouping the trace logarithm and using

$$\frac{\mu_p^{2r}}{1-\mu_p^r} = \sum_{k\geq 2} \mu_p^{kr}$$

gives (2); in particular the lower index is forced to be 2. There are at most $2^n/n$ primitive length-n orbits and $|\mu_p| \leq q^n$, so the sum of absolute values of all product factors is bounded by

$$\sum_{n\geq 1} \frac{|u|^n 4^{-n}}{n(1-q^n)},$$

which converges for $|u| < 4$. $\square$

Exact receipt. Möbius inversion gives primitive counts $2, 1, 2, 3, 6, 9$ for $n = 1, \ldots, 6$. Exact arithmetic in $\mathbb{Q}[z]/(F^n - z)$ gives the traces and Fredholm coefficients below; $c_0 = 1$ and $D_2(u) = \sum c_j u^j$.

n	$\operatorname{Tr} \mathcal{L}_2^n$	c_n
1	$\frac{1}{12}$	$-\frac{1}{12}$
2	$\frac{7}{720}$	$-\frac{1}{720}$
3	$\frac{239}{257472}$	$-\frac{1}{1287360}$
4	$\frac{1255703}{13810694400}$	$-\frac{1}{2057793465600}$
5	$\frac{235072563599}{26491011084499968}$	$-\frac{1}{262890767297555958689280 0}$
6	$\frac{655398850662090042240821783}{756396676602907446734765701632000}$	$-\frac{1}{2145321764151480887652914286846712748095922688000}$

The recurrence is $c_n = -n^{-1} \sum_{j=1}^n c_{n-j} \operatorname{Tr} \mathcal{L}_2^j$. The six rows are a replay prefix, not a theorem cutoff.

Negative control, progress, and boundary. For $z^2 - 2$ the same $\mathbb{D}_4$ construction fails because the branch point -2 lies inside the disk; other owner spaces are not excluded. Relative to finite polynomial matrices, real count models, and Möbius word matrices in this series, the advance is one nonlinear complex polynomial with intrinsic branches, unconditional nuclearity, all-period exhaustion, and a stability-weight ladder. No external novelty claim for general Ruelle theory is made.

The strict Route-A tuple is (A1_WEAK, A2_FAIL, A3_FAIL, A4_FAIL); Route B is false. There is no prime-like target correspondence, target divisor or functional equation, arithmetic/local data, Euler factor, root number, automorphy, natural unitary or self-adjoint quantization, or Hilbert–Pólya claim.

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